Topic 4 — Digestion, Respiration and Excretion

Covers syllabus (click to expand)

Digestive System Anatomy (Mouth, Teeth, Salivary Glands, Stomach, Liver, Gall Bladder, Pancreas, Small & Large Intestine) | Dental Formula & Tooth Structure (Enamel) | Digestive Enzymes & Juices (Ptyalin, Pepsin, Rennin, Bile, Trypsin, Lipase, Succus Entericus) | Digestion of Carbohydrates, Proteins, Fats, Starch & Cellulose | Absorption Mechanics (Villi, Lacteals, Chylomicrons) | Respiratory System (Airway Anatomy, Larynx, Trachea, Alveoli) | Breathing Mechanics (Diaphragm, Intercostal Muscles) & Lung Volumes (Tidal, Vital Capacity, Residual) | Gas Transport (O_2 Oxyhaemoglobin, CO_2 Bicarbonate, Bohr Effect) & Carbon Monoxide Poisoning | Cellular Respiration (Glycolysis, Krebs Cycle, ETS, ATP Yield, Lactic Acid Debt) | Excretory System Anatomy (Kidney, Nephron Structure, Malpighian Body, Tubules) | Urine Formation (Ultrafiltration, Selective Reabsorption, Tubular Secretion) | Counter-Current Mechanism & Hormonal Regulation (ADH, Aldosterone, RAAS, ANF) | Nitrogenous Wastes (Ammonotelic, Ureotelic, Uricotelic) | Excretory Disorders (Renal Calculi, Uremia, Gout, Dialysis)

Sources baked in: NCERT Class 10 & 11 (Life Processes, Digestion and Absorption, Breathing and Exchange of Gases, Excretory Products and Elimination), Lucent General Science (Physiology & Enzymes), UPPCS Prelims PYQs (2018–2025) and UKPCS Prelims **Weight:** ★★★ High — heavily tested: site of initiation of protein digestion, stomach vs pancreatic enzymes, bile function, lung volumes, gas transport forms, nephron reabsorption sites, and nitrogenous waste types **Last verified:** September 2026

Current Affairs

Year	Event / Fact	Must-Score Fact
2024	UKPCS tested agent class utilized as an **immunosuppressor in kidney transplantation**.	Keyed answer: **Monoclonal antibodies** (e.g. Basiliximab / Daclizumab) and calcineurin inhibitors (Cyclosporine A), preventing allograft organ rejection; antibiotics do not prevent graft rejection.
Still Current	**Kidney stone composition in India**: Over **80% of urinary calculi** are composed of insoluble **Calcium Oxalate (CaC_2O_4) crystals**, followed by calcium phosphate and uric acid.	Dietary facts: Foods rich in oxalates (tomatoes, spinach, beetroot, chocolate) accelerate stone formation in genetically prone individuals.
Still Current	Medical device fact: **Haemodialysis machine ("Artificial Kidney")** operates on the physical principle of **Dialysis across a semi-permeable cellophane membrane**.	Patient's blood is drawn from an artery, mixed with anticoagulant **Heparin**, filtered against dialyzing fluid, rewarmed, and reinfused with **Anti-heparin** into a vein.

4.1 Digestive Physiology: From Mouth to Absorption

- **The Human Mouth & Mechanical Mastication**:
- Food is ingested into the buccal cavity, where it is mechanically masticated by teeth and lubricated by saliva to form a swallowed mass called a **Bolus**.
- **Human Dentition Characteristics**:
    - **Thecodont**: Every tooth is embedded in a deep bony jaw socket (alveolus).
    - **Diphyodont**: Two sets of teeth during lifetime: deciduous / milk teeth (20 teeth) replaced by permanent adult teeth (32 teeth).
    - **Heterodont**: Four morphologically and functionally distinct types of teeth:
    - **Incisors (I)**: Chisel-shaped, for biting and cutting food.
    - **Canines (C)**: Dagger-shaped, sharp, for tearing meat (prominent in carnivores; absent in rodents \rightarrow toothless gap termed **Diastema**).
    - **Premolars (PM)**: Bicuspid, for grinding and chewing food (absent in milk dentition).
    - **Molars (M)**: Broad, multicuspid, for crushing and grinding food; the 3rd molar on each side is the **Wisdom Tooth** (erupts at age 17–25; vestigial).
- **Salivary Glands**: Three pairs of exocrine salivary glands secrete 1.0–1.5 litres of saliva per day:
    1. **Parotid Glands**: Largest pair, situated below and in front of the ears; opens via **Stensen's duct**; infection by Paramyxovirus causes painful swelling called **Mumps**.
    2. **Submandibular / Submaxillary Glands**: Located at the angles of the lower jaw; opens via **Wharton's duct**; secretes the majority of resting saliva.
    3. **Sublingual Glands**: Smallest pair, situated under the tongue; opens via **Ducts of Rivinus**. - Saliva contains **Ptyalin (Salivary Amylase)** which digests cooked starch into maltose, **Lysozyme** (bactericidal enzyme cleaving bacterial cell walls), secretory **IgA**, and bicarbonate buffer (pH 6.8).
- **Gastric Digestion in the Stomach**:
- The stomach is a J-shaped muscular organ with 4 regions: Cardiac, Fundus, Body, and Pylorus. Guarded by the **Gastro-oesophageal sphincter** at the entry and the **Pyloric sphincter** at the exit.
- Gastric mucosa contains branched tubular **Gastric Glands** possessing three major cell types:
    1. **Mucous Neck Cells**: Secrete protective alkaline mucus.
    2. **Peptic / Chief / Zymogenic Cells**: Secrete inactive proenzymes **Pepsinogen** and **Prorennin**.
    3. **Parietal / Oxyntic Cells**: Secrete **Hydrochloric Acid (HCl)** and **Castle's Intrinsic Factor** (indispensable for Vitamin B12 absorption in the ileum).
- Functions of Gastric HCl: Creates a strongly acidic environment (pH 1.5–2.0); kills ingested food-borne bacteria; decalcifies bone fragments; converts inactive pepsinogen into active **Pepsin**.
- **Pepsin** cleaves internal peptide bonds of dietary proteins, converting complex proteins into soluble **Proteoses and Peptones**.
- Food is churned into an acidic, semi-fluid pulpy mass called **Chyme**.
- **Duodenal Digestion: Liver, Bile & Pancreas**:
- Acid chyme enters the C-shaped **Duodenum** through the pyloric valve; triggers secretion of two local enterogastrone hormones:

  - **Secretin**: Discovered by **Bayliss and Starling (1902)** (the **first discovered hormone in history**); stimulates pancreatic secretion of watery bicarbonate juice to neutralise gastric acid.
  - **Cholecystokinin (CCK / Pancreozymin)**: Stimulates gall bladder contraction to eject bile, and stimulates pancreatic secretion of digestive enzymes.

- **The Dual Pancreas**:
  - Endocrine Islets of Langerhans (1–2% mass): Insulin and Glucagon (Topic 3).
  - Exocrine Acinar Tissue (98% mass): Secretes complete pancreatic juice into the duodenum via the **Pancreatic Duct (Duct of Wirsung)**, joining the common bile duct to form the **Hepato-pancreatic Ampulla (Ampulla of Vater)**, guarded by the **Sphincter of Oddi**.

- **Intestinal Absorption & The Hepatic Portal Axis**:
- The small intestine (Jejunum and Ileum) completes digestion via brush-border enzymes of the **Succus Entericus**.
- Monosaccharides (glucose, galactose) and amino acids are actively transported across villus enterocytes into local blood capillaries; coalesce into the **Hepatic Portal Vein**, which transports newly absorbed nutrients directly to the **Liver** for metabolic sorting, storage, and detoxification before systemic circulation.
- Long-chain fatty acids and fat-soluble vitamins cannot enter blood capillaries directly; packaged into protein-jacketed **Chylomicrons**, absorbed into central lymphatic **Lacteals**, and drained via the **Thoracic Lymphatic Duct** into the left subclavian vein.

Bilingual Terminology -- Digestion, Respiration and Excretion

English Term	Hindi Term	Key Anchor
Digestion (पाचन)	पाचन	Breaking down food into absorbable molecules
Alimentary Canal (आहार नाल)	आहार नाल / पाचन नाल	Mouth to anus; complete digestive tract
Saliva (लार)	लार	Secretion of salivary glands; contains **salivary amylase** (ptyalin)
Stomach (आमाशय)	आमाशय	Secretes gastric juice (HCl + pepsin + mucin)
Small Intestine (छोटी आंत)	क्षुद्रांत्र	Main site of digestion and absorption; duodenum, jejunum, ileum
Large Intestine (बड़ी आंत)	वृहदांत्र	Water absorption; formation of faeces; colon, rectum
Liver (यकृत)	यकृत	Largest gland; secretes bile; detoxification; glycogen storage
Pancreas (अग्न्याशय)	अग्न्याशय	Secretes pancreatic juice (amylase, lipase, trypsinogen); also insulin/glucagon
Bile (पित्त)	पित्त	Emulsifies fats; secreted by liver; stored in gallbladder

English Term	Hindi Term	Key Anchor
Villi (आंत्र अंकुर)	रसांकुर	Finger-like projections in small intestine; increase surface area for absorption
Respiration (श्वसन)	श्वसन	Process releasing energy from food; cellular respiration = aerobic/anaerobic
Aerobic Respiration (वायवीय श्वसन)	वायवीय श्वसन	Uses O2; produces CO2 + H2O + 38 ATP per glucose
Anaerobic Respiration (अवायवीय श्वसन)	अवायवीय श्वसन	No O2; produces lactic acid (muscles) or ethanol+CO2 (yeast)
Alveoli (वायुकोश)	वायुकोश	Tiny air sacs in lungs; site of gas exchange (O2 in, CO2 out)
Diaphragm (मध्यपट)	मध्यपट	Dome-shaped muscle below lungs; main breathing muscle
Excretion (उत्सर्जन)	उत्सर्जन	Removal of metabolic waste products from the body
Kidney (वृक्क)	वृक्क / गुर्दा	Primary excretory organ; filters blood to form urine
Nephron (नेफ्रोन)	वृक्काणु / नेफ्रोन	Structural and functional unit of kidney; ~1 million per kidney
Urea (यूरिया)	यूरिया	Main nitrogenous waste in mammals; formed in liver
Uric Acid (यूरिक अम्ल)	यूरिक अम्ल	Nitrogenous waste in birds, reptiles, insects (less water loss)
Dialysis (डायलिसिस)	डायलिसिस	Artificial kidney function; filters blood when kidneys fail

4.2 Respiratory Mechanics & Gas Transport Dynamics

- **The Respiratory Pathway Anatomy**:
- External nostrils \rightarrow Nasal cavity (lined by vascular pseudostratified ciliated columnar epithelium and olfactory Schneiderian membrane) \rightarrow Pharynx (common passage for food and air) \rightarrow Larynx (guarded by the elastic cartilaginous Epiglottis) \rightarrow Trachea (C-shaped cartilage rings) \rightarrow Primary Bronchi \rightarrow Secondary/Tertiary Bronchi \rightarrow Terminal Bronchioles \rightarrow Respiratory Bronchioles \rightarrow Alveolar Ducts \rightarrow **Alveoli**.
- **The Alveolar Respiratory Membrane**:
  - Ultra-thin diffusion barrier (<0.5\text{ microns} thick) composed of: (a) Alveolar type-I squamous epithelium, (b) Epithelial basement membrane, (c) Capillary basement membrane, and (d) Capillary endothelial wall.
  - Alveolar type-II cells secrete **Pulmonary Surfactant (Dipalmitoyl lecithin)**; lowers surface tension of alveolar fluid, preventing lung collapse during expiration; premature neonates lacking surfactant develop fatal **Infant Respiratory Distress Syndrome (IRDS)**.
- **Physical Chemistry of Gas Transport**:
- **Oxygen (O_2) Exchange**:

    - Alveolar partial pressure of oxygen (pO_2) is high (~104\text{ mm Hg}), while deoxygenated venous capillary pO_2 is low (~40\text{ mm Hg}); O_2 diffuses rapidly into blood.
    - Haemoglobin consists of globin protein and 4 haem iron groups; binds oxygen cooperatively, producing a characteristic **Sigmoidal Oxyhaemoglobin Dissociation Curve**.
    - **Right Shift of Dissociation Curve (Facilitates O_2 Release at Tissues)**: Caused by elevated pCO_2, elevated temperature, elevated 2,3-diphosphoglycerate (2,3-DPG), and decreased blood pH / acidosis (**The Bohr Effect**).
    - **Left Shift of Dissociation Curve (Increases O_2 Affinity)**: Caused by low pCO_2, high pH (alkalosis), and lower temperature (as seen in alveolar capillaries).
- **Carbon Dioxide (CO_2) Exchange**:
    - Tissue pCO_2 is high (~45\text{ mm Hg}); diffuses into blood capillaries where pO_2 is lower (~40\text{ mm Hg}).
    - **Bicarbonate Mechanism (70\%)**: CO_2 + H_2O \xrightleftharpoons[\text{Anhydrase}]{\text{Carbonic}} H_2CO_3 \rightleftharpoons H^+ + HCO_3^-
    - As HCO_3^- ions accumulate inside erythrocytes, they diffuse down their concentration gradient into blood plasma; to preserve electrical neutrality, chloride ions (Cl^-) diffuse from plasma into the erythrocyte — this phenomenon is the **Chloride Shift / Hamburger Phenomenon**.
- **Cellular Respiration Pathways & Energetics**:
- **Phase 1: Glycolysis (EMP Pathway)**:
    - Occurs in the **cytoplasm**; anaerobic (does not require O_2); common to both aerobic and anaerobic organisms.
    - Converts 1 molecule of Glucose (6C) into 2 molecules of Pyruvate (3C); yields net **2 ATP** (substrate-level phosphorylation) and **2 NADH**.
- **Phase 2: Link Reaction (Oxidative Decarboxylation)**:
    - Pyruvate enters the **mitochondrial matrix**; converted to Acetyl-CoA by Pyruvate Dehydrogenase, releasing CO_2 and yielding 2 NADH.
- **Phase 3: Krebs Cycle (Citric Acid / TCA Cycle)**:
    - Occurs in the **mitochondrial matrix**; discovered by Sir Hans Krebs (Nobel 1953).
    - Acetyl-CoA (2C) combines with Oxaloacetate (4C) to form Citrate (6C); cyclic oxidation releases CO_2, yielding **2 ATP / GTP**, **6 NADH**, and **2 FADH2**.
- **Phase 4: Oxidative Phosphorylation (ETS)**:
    - Inner mitochondrial membrane cristae; electrons from NADH and FADH2 pass through cytochromes to molecular oxygen (terminal electron acceptor, forming H_2O); drives proton gradient pumping through F_0-F_1 complexes to generate ATP.
    - Total net yield: **36 to 38 ATP per molecule of glucose**.

4.3 Renal Excretion, Nephron Mechanics & Water Balance

- **Gross Anatomy of the Human Urinary System**:

- Pair of bean-shaped reddish-brown kidneys located retroperitoneally between T12 and L3 vertebrae; right kidney is positioned slightly lower than left due to the massive liver above it.
- External notch called the **Hilum** admits renal artery, renal vein, nerves, and the **Ureter**.
- Internally divided into an outer pale **Cortex** and inner dark **Medulla** consisting of 8–18 conical **Renal Pyramids** projecting into minor and major calyces, opening into the funnel-shaped **Renal Pelvis**.
- **Ultrastructure of the Nephron**:
- Two primary anatomical divisions:
    1. **Renal Corpuscle (Malpighian Body)**: Located in the cortex. Composed of the **Glomerulus** (a high-pressure tuft of non-anastomosing capillaries supplied by a wide **Afferent Arteriole** and drained by a narrower **Efferent Arteriole**) enclosed within **Bowman's Capsule**. - Inner visceral wall of Bowman's capsule is lined by specialized epithelial cells called **Podocytes**, whose pedicels interdigitate to leave narrow **Filtration Slits (Slit Pores)** (~25\text{ nm} wide).
    2. **Renal Tubule**: - **Proximal Convoluted Tubule (PCT)**: Located in cortex; brush-border microvilli. - **Loop of Henle**: Hairpin loop dipping into the medulla (long in juxtamedullary nephrons). - **Distal Convoluted Tubule (DCT)**: Located in cortex. - **Collecting Duct**: Drains multiple DCTs; descends through medullary pyramids to the duct of Bellini.

- **The Three Stages of Urine Formation**: 1. **Glomerular Ultrafiltration**:
    - Driven by **Net Filtration Pressure (NFP)**: $\text{NFP} = \text{Glomerular Hydrostatic Pressure (GHP \sim 60)} - [\text{Colloid Osmotic Pressure (BCOP \sim 32)} + \text{Capsular Hydrostatic Pressure (CHP \sim 18)}] \approx \mathbf{10\text{ mm Hg}}
    - Produces **180 litres of protein-free glomerular filtrate per day** (GFR = 125\text{ mL/min}). 2. **Selective Tubular Reabsorption**:
    - Over **99\% of the 180-litre filtrate is reabsorbed** back into peritubular capillaries, leaving only **\sim 1.5\text{ litres} of urine** per day.
    - Glucose, amino acids, and Na^+ are reabsorbed via high-affinity active transport; water follows passively via osmosis. 3. **Tubular Secretion**:
    - Active transport of waste ions (H^+, K^+, NH_4^+), creatinine, and drugs (penicillin, sulphonamides) from peritubular capillaries directly into the tubular lumen; vital for maintaining blood pH (7.35–7.45) and electrolyte balance.

- **Counter-Current Multiplier & Concentrator**:
- The parallel counter-directional flow of tubular fluid in the descending and ascending limbs of Henle's loop, combined with counter-current blood flow in the **Vasa Recta** capillaries, establishes a steep hyperosmotic interstitial gradient in the renal medulla (from 300\text{ mOsm/L} in cortex to \mathbf{1200\text{ mOsm/L}} in deep renal papilla).
- Enables humans to concentrate urine up to **4 times higher** than normal blood plasma osmolarity (~1200\text{ mOsm/L}), conserving critical body water.
- **Hormonal Feedback Loops Governing Excretion**:
- **Renin-Angiotensin-Aldosterone System (RAAS)**:
    - Trigger: Drop in GFR, renal blood flow, or arterial pressure.
    - **Juxtaglomerular Cells (JG Cells)** of afferent arterioles secrete the proteolytic enzyme **Renin**.
    - Renin cleaves plasma **Angiotensinogen** (made by liver) \rightarrow **Angiotensin-I**.

    - **Angiotensin Converting Enzyme (ACE)** in lung capillaries converts Angiotensin-I \rightarrow **Angiotensin-II**.
    - Angiotensin-II is a potent arterial vasoconstrictor that elevates blood pressure and stimulates the adrenal cortex to secrete **Aldosterone**.
    - Aldosterone stimulates active Na^+ and water reabsorption in the DCT and collecting duct, restoring blood volume.

- **Antidiuretic Hormone (ADH / Vasopressin)**:
    - Trigger: Hypothalamic osmoreceptors detect dehydration (blood osmolarity >300\text{ mOsm/L}).
    - Synthesised in hypothalamus, released from posterior pituitary; acts on collecting ducts to promote rapid water reabsorption.

- **Atrial Natriuretic Factor (ANF)**:
    - Secreted by myocardial cells of cardiac atria when blood pressure is excessively elevated; acts as a powerful vasodilator and stimulates excretion of sodium (Na^+) and water, directly opposing RAAS.

One-Liner Revision -- Digestion, Respiration and Excretion

#	Fast Fact
1	Digestion starts in the **mouth** (salivary amylase/ptyalin digests starch)
2	Stomach secretes = HCl + Pepsin (protein digestion) + Mucin (protects lining)
3	Liver = largest **gland** of the body; also largest internal organ
4	Bile secreted by = **liver**; stored in = **gallbladder**
5	Main site of digestion and absorption = **Small intestine** (duodenum/jejunum/ileum)
6	Large intestine main function = **water absorption**; no digestion
7	Longest part of alimentary canal = **Small intestine** (~6-7 metres)
8	Pancreas = both **exocrine** (digestive enzymes) and **endocrine** (insulin/glucagon)
9	Aerobic respiration: Glucose + O2 = CO2 + H2O + **38 ATP**
10	Anaerobic in muscles = **lactic acid** (causes muscle cramp/fatigue)
11	Anaerobic in yeast = **ethanol + CO2** (fermentation; used in brewing)
12	Breathing rate (adults) = 15-18 breaths/min at rest
13	Tidal volume = air breathed in/out in one normal breath (~500 mL)
14	Vital capacity = maximum air exhaled after maximum inhalation (~4.5 L)
15	Gas exchange in lungs = in **alveoli** (O2 diffuses in; CO2 diffuses out)

#	Fast Fact
16	Haemoglobin carries **O2** as oxyhaemoglobin; also carries some CO2
17	CO poisoning = CO binds Hb **240x more tightly** than O2 (cherry-red colour)
18	Mammals excrete **Urea** (ureotelic); Birds/reptiles excrete **Uric acid**
19	Fish excrete **Ammonia** (ammonotelic; highly toxic; needs lots of water)
20	Nephron = functional unit of kidney; Bowman's capsule + tubules + loop of Henle
21	Urine formation: Filtration + Selective reabsorption + Tubular secretion
22	Normal urine output = ~1-1.5 L/day; urine pH ~6 (slightly acidic)
23	Kidney stone = **calcium oxalate** (most common); treated by lithotripsy
24	Dialysis = used when kidneys fail; hemodialysis filters blood through machine
25	Skin is also an excretory organ = excretes sweat (water, NaCl, urea)
26	Lungs excrete = CO2 and water vapour
27	Liver excretes = bile pigments (bilirubin from dead RBCs)
28	ADH (Anti-diuretic hormone) = controls water reabsorption in kidney tubules
29	Jaundice = yellow skin/eyes due to accumulation of **bilirubin**
30	Appendix = vestigial organ in large intestine; removed in appendicitis